

Task 1-Student Grade Tracker

Source Code

```
import java.util.ArrayList;
import java.util.Scanner;

public class StudentGradeTracker {
    public static void main(String[] args) {
        Scanner scanner = new Scanner(System.in);

        ArrayList<String> studentNames = new ArrayList<>();
        ArrayList<Integer> studentGrades = new ArrayList<>();

        System.out.print("Enter number of students: ");
        int numStudents = scanner.nextInt();
        scanner.nextLine(); // Consume newline

        for (int i = 0; i < numStudents; i++) {
            System.out.print("Enter student name: ");
            String name = scanner.nextLine();
            studentNames.add(name);

            System.out.print("Enter grade for " + name + ": ");
            int grade = scanner.nextInt();
            scanner.nextLine(); // Consume newline
            studentGrades.add(grade);
        }

        // Calculate average, highest, and lowest
        int total = 0;
        int highest = Integer.MIN_VALUE;
        int lowest = Integer.MAX_VALUE;

        for (int grade : studentGrades) {
            total += grade;
            if (grade > highest) highest = grade;
            if (grade < lowest) lowest = grade;
        }

        double average = (double) total / studentGrades.size();

        // Display results
        System.out.println("\n--- Grade Summary ---");
        for (int i = 0; i < studentNames.size(); i++) {
            System.out.println(studentNames.get(i) + ": " + studentGrades.get(i));
        }

        System.out.println("Average Grade: " + average);
    }
}
```

```
System.out.println("Highest Grade: " + highest);  
System.out.println("Lowest Grade: " + lowest);
```

```
scanner.close();
```

```
}  
}
```